

CONTACT

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NANDITHA RAO

SOFTWARE DEVELOPER

SKILLS

Languages

- Typescript
- Java
- JavaScript
- HTML + CSS
- Python
- PHP
- *Past exposure to: C, C++, C#, ASP.NET*

Frameworks

- Spring Boot
- Bootstrap
- Node.js
- Next.js (early exposure)
- GraphQL (early exposure)

Backend & APIs

- REST
- Swagger/OpenAPI
- Postman
- YAML

Tools & Platforms

- Docker
- Git/GitHub
- Jira
- Linux

Databases

- MySQL

Other

- CI/CD collaboration
- Data modelling
- Cross-team API integration

EDUCATION

MS, Computer Science

Western University
2017 – 2019
Research Assistant | Teaching Assistant

MCA

Mount Carmel College, India
2012 – 2015
Received Award for Academic Excellence

PROFESSIONAL OVERVIEW

Full-stack Software Developer with 4+ years of professional experience designing, modernizing, and maintaining enterprise-grade backend and web applications. Specialized in TypeScript, Java/Spring Boot, REST API architecture, SQL, and high-performance backend systems. Proven ability to replace legacy platforms with scalable, maintainable solutions, build reusable libraries adopted across teams, and own end-to-end technical design and implementation. Recognized for clarity in architecture, reliability, and delivering maintainable systems across enterprise environments.

EXPERIENCE

Software Developer II

Cineplex Digital Media (CDM), London, ON | Nov 2021 – Nov 2025

(Position concluded as part of company-wide restructuring following acquisition)

- Served as one of the core developers responsible for **rebuilding CDM's backend** from legacy, undocumented SOAP systems into a clean, scalable, **TypeScript-based API ecosystem** handling 50+ domain entities across multiple relational databases.
- Owned the **end-to-end lifecycle** for each service, including API contract design, structured JSON response modelling, data validation, service implementation, reusable Node and Java library creation, and deployment testing in Docker.
- Designed and implemented **complex data-fetching/modifying routes**:
 - GET routes supported multiple query patterns (for full objects, "mini" objects or ID-only arrays while filtering subsets based on nearly all properties of said complex objects), enabling **flexible, high-performance access** despite highly normalized, interdependent database schemas.
 - POST/PUT/PATCH operations involved multi-step transactional SQL queries executed in strict sequences to **maintain data integrity across interdependent entities**.
- Built **client-specific APIs** with authentication and access control, alongside general-purpose services for internal tooling and system-wide workflows, ensuring consistent behavior and maintainability across the backend while safeguarding sensitive enterprise data.
- Developed and maintained reusable **Node (TypeScript)** and **Java** libraries with comprehensive **unit tests**, ensuring correctness, reliability, and maintainability across multiple front-end teams.

PUBLICATIONS

Analysis of the Effect of QoS on Video Conferencing QoE, *IWCMC, 2019*

Classrooms in 2025: Technology vs. Human Pedagogy, *International Conference on Higher Education, 2015*

ADDITIONAL TECHNICAL EXPERIENCE

Bengaluru, India | 2015 – 2016

- **Indian Space Research Organisation (ISRO)**

Developed a telemetry data decoder and visualization tool for satellite health monitoring, creating secure data pipelines from satellite → ground application → user interface.

- **Indian Oil Corporation**

Supported enterprise software operations, generated domain reports, resolved system bugs, and improved reliability of distributor business systems. Early work involved troubleshooting cascading issues within low-tolerance environments and improving stability of legacy systems.

- Centralized shared logic and common operations in dedicated modules to **reduce duplication**, **enforce coding standards**, and **accelerate onboarding for new developers**; mentored team members and conducted code reviews to maintain engineering quality.
- Maintained modular, responsibility-separated codebases that mirrored **microservice principles**, enabling rapid addition of new services while isolating impact from existing functionality.
- Optimized SQL queries and data access patterns to improve API response times, significantly outperforming legacy SOAP services.
- **Collaborated** closely with front-end engineers to align API responses with user-facing features and enterprise client workflows, serving thousands of requests per day across multiple enterprise clients.
- Created Docker-based local testing environments; extensively used **Postman** and **Swagger/OpenAPI** for validation, debugging, and documentation.
- Proposed future modernization initiatives using **Next.js** and **GraphQL**, conducting technical scoping, stakeholder alignment, and early proof-of-concept implementation.
- Independently developed, maintained, refactored, and improved core services after the senior lead was laid off, ensuring ongoing reliability and performance for enterprise clients such as **RBC**, **Dairy Queen**, **Cadillac Fairview**, and others.
- Experimented with AI-assisted development using Copilot; built an interactive 3D Rubik's Cube prototype during a one-hour internal engineering demo to showcase rapid prototyping and AI-aided code generation.
- Transitioned from onsite to full-time remote post-pandemic along with the rest of the team.

Application Engineer

Starlim North America (SNA), London, ON | Feb 2020 – Oct 2021

- Developed, maintained, and enhanced Starlim's internal web-based ERP system and supporting applications using Java, JavaScript, HTML/CSS and SQL.
- Built custom in-house tools and services supporting core manufacturing, HR, and administrative operations.
- Provided 2nd and 3rd level technical support for ERP users across multiple departments, ensuring consistent system uptime and data accuracy.
- Designed Starlim's internal mobile application to deliver essential functions to employees.

Research Associate

Western University (UWO), London, ON | Apr 2019 – Feb 2020

- Supervised and trained new research team members studying correlations between network performance metrics (QoS) and user experience metrics (QoE).
- Conducted experiments and data analysis contributing to an autonomous system for predicting QoE based on real-time network metrics.